Gym Management System

Database Design & Analytics Project

Project Team

Thomas Nady Shawky Armanious

Abdelrahman Hossam Salah Abdelhakim

Gomana Saeed Mohamed

Silvana Mohsen

Rahma Ramadan Ahmed Mahmoud

Alaa Osama Abdelfattah Omar

Supervised by

Eng. Menna Tarek

Table of Contents

Introduction

Project Objective

- Analyze the Gym Management System data to monitor member activity, examine workout behavior, and evaluate branch performance.
- Identify peak usage times, member distribution patterns, and the impact of subscription type and gym type on overall performance.

Why This Analysis Matters

A clear understanding of member behavior and branch performance enables gym management to make more informed decisions around marketing, pricing, and overall service quality.

Dataset Summary

The data source consists of 6 tables after applying normalization:

Table	Source File
Member	users_data.csv
Gym_Branch	gym_locations_data.csv
Facility	Derived from gym_locations_data.csv
Gym_Facility	Junction table
Subscription_Plan	subscription_plans.csv
Workout_Session	checkin_checkout_history_updated.csv

Used Tools

Cleaning & Analysis

- MySQL
- Excel & Power Query
- Python

Visualization

- Power BI

Data Cleaning

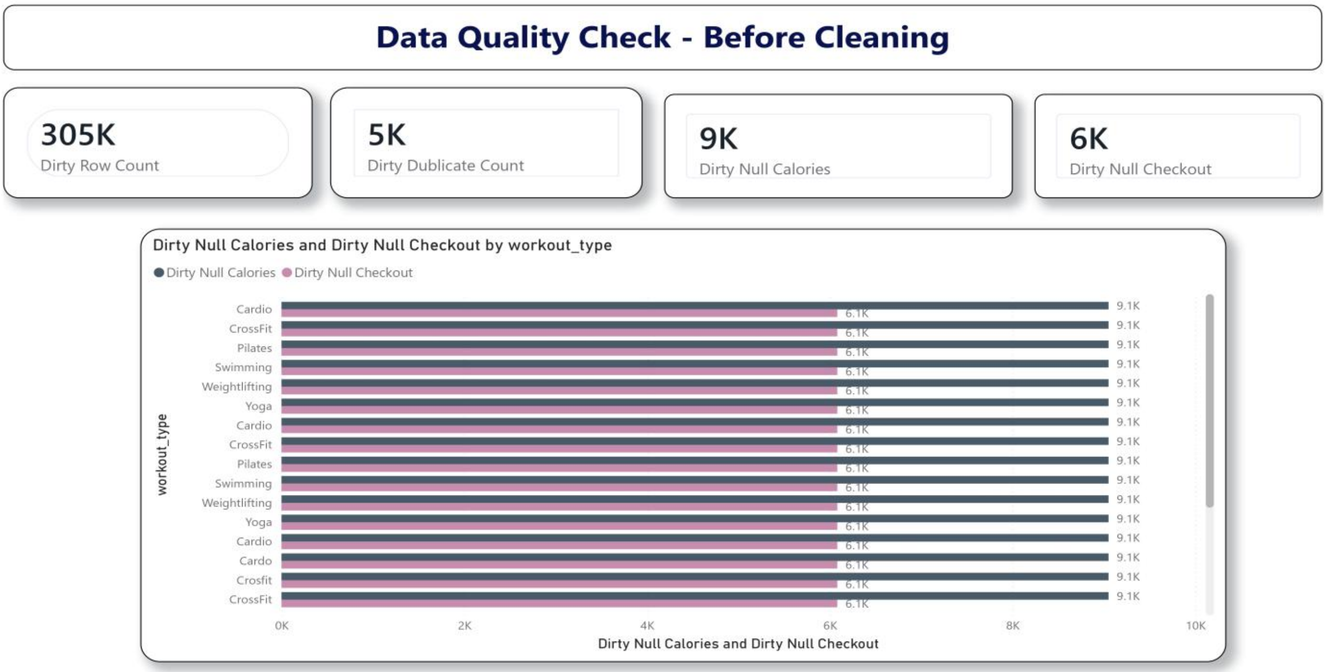
The Workout_Session table exhibited the following data quality issues (300,000 → 304,500 rows once duplicates are counted):

Issue	Affected Column	Impact %	Cleaning Stage
Missing values	calories_burned	3.0%	Python
Missing values	checkout_time	2.0%	Excel
Duplicate rows	All columns	1.5%	SQL
Inconsistent casing / whitespace	workout_type	5.0%	SQL
Spelling typos (Yogaa, Crosfit)	workout_type	2.0%	Excel
Inconsistent date format	checkin_time	3.0%	Excel
Outliers (negative or 9999+)	calories_burned	1.0%	Python
Logical error (checkout before checkin)	checkout_time	1.0%	SQL
Leading/trailing whitespace	user_id	2.0%	SQL

Cleaning was carried out in three stages: Stage 1 (partial cleaning in SQL) → Stage 2 (Excel / Power Query) → Stage 3 (fully clean in Python).

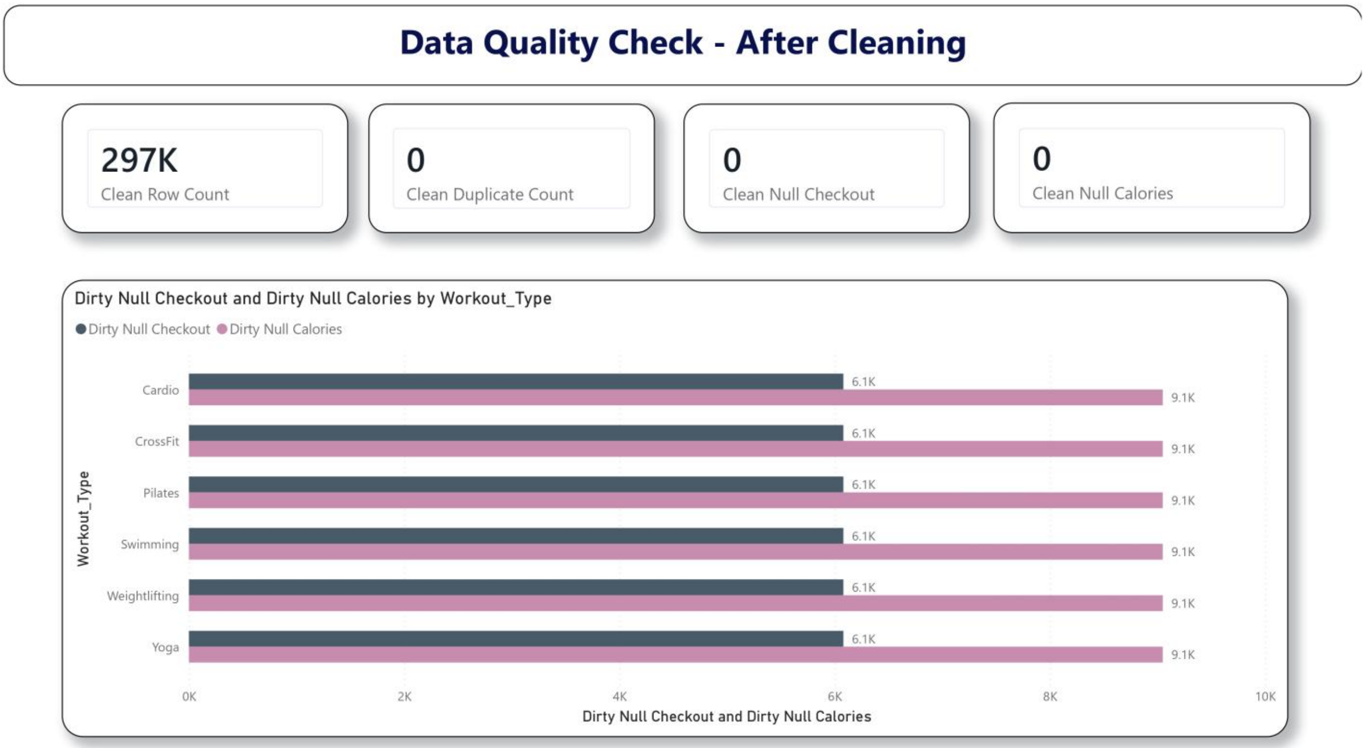
Data Quality Check — Before & After Cleaning

Before Cleaning



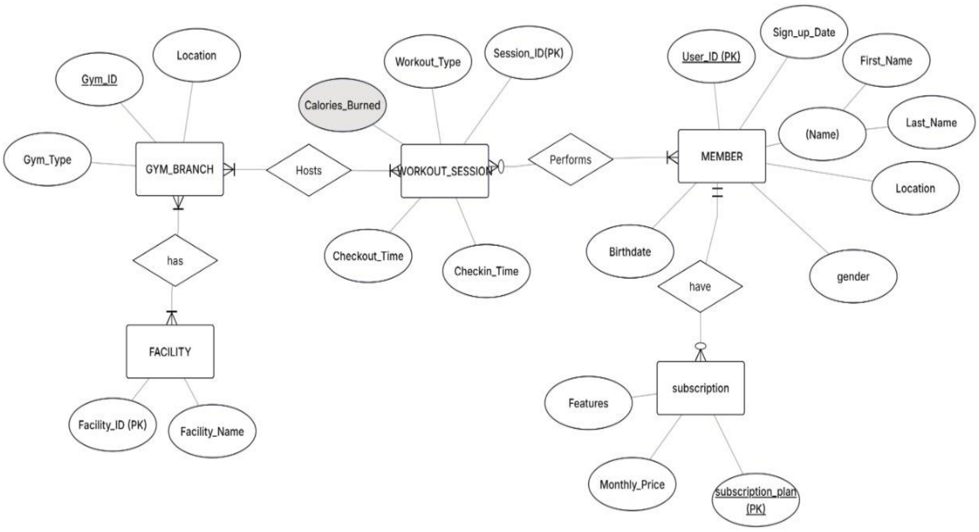
305K rows, 5K duplicates, 9K null calorie values, 6K null checkout values

After Cleaning



297K clean rows, zero duplicates, zero null values

Data Model (ERD)

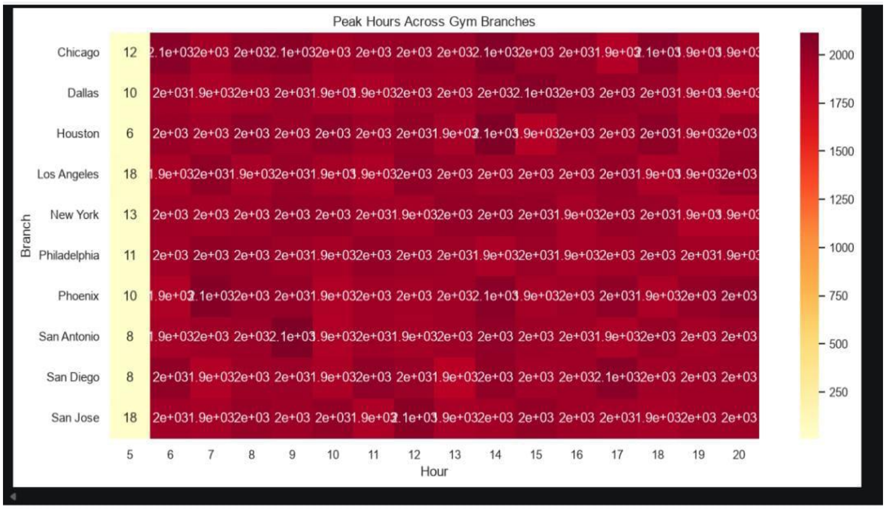


Analysis & Insights

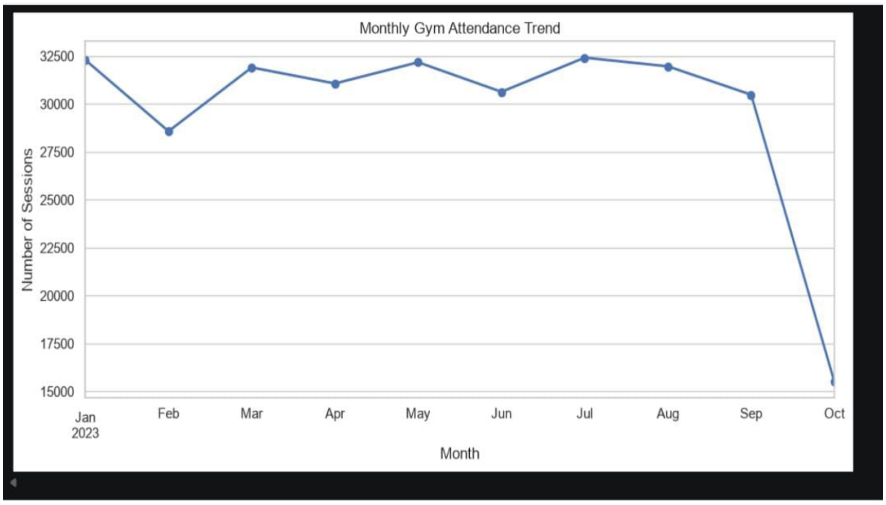
Branch Insights

- Member distribution is relatively balanced across the five branches.
- Peak usage hours consistently occur during daytime across all branches.
- The monthly attendance trend is generally stable, with a notable drop in the final month attributable to incomplete data logging.
- Average calories burned are comparable across the Budget, Standard, and Premium gym types.

Peak Hours Across Gym Branches



Monthly Gym Attendance Trend

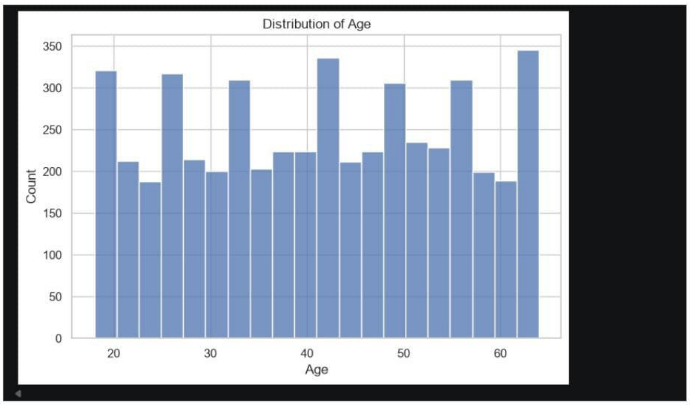


Member Insights

- Member ages are fairly evenly distributed between 18 and 64 years old.
- Gender distribution: Male 46.6%, Female 43.6%, Non-binary 9.8%.
- Calories burned are similar across the Basic, Student, and Pro subscription plans.
- The correlation between age and calories burned is approximately 0, indicating no meaningful relationship between the two variables.

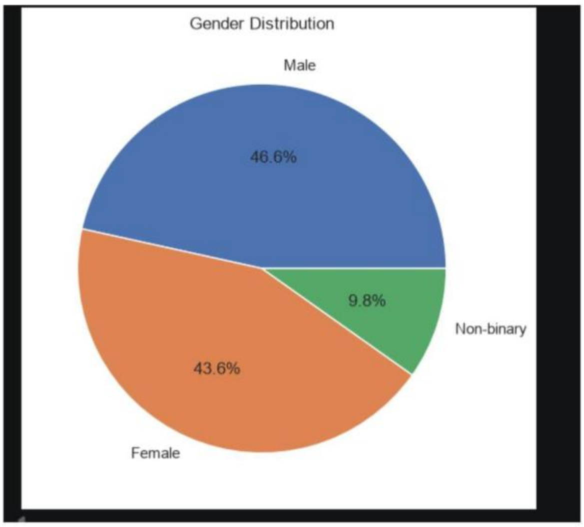
Distribution of Age

3. Histogram of ages



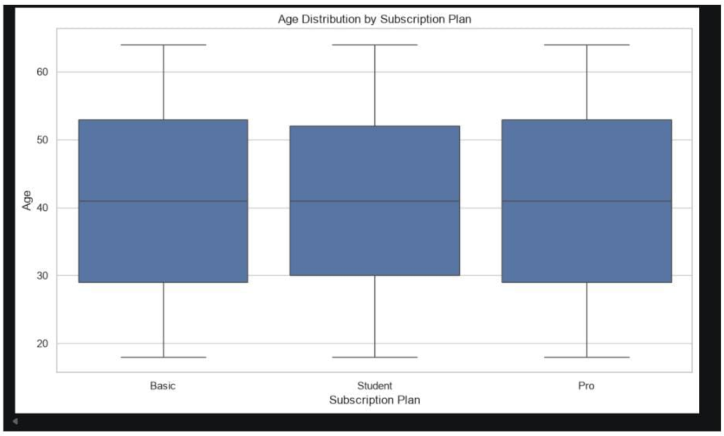
Gender Distribution

4. Pie chart for gender distribution (common names: gender, sex)



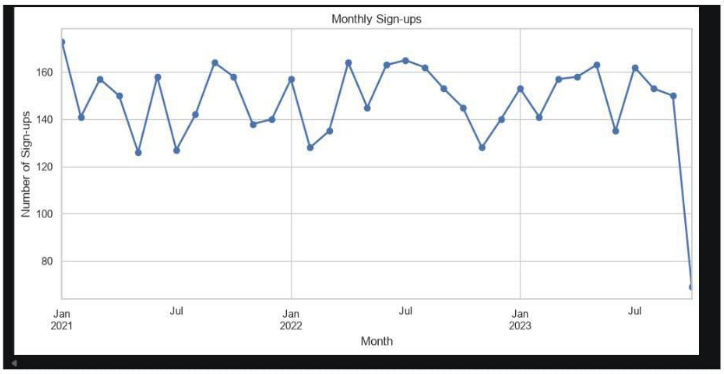
Calories Burned by Subscription Plan (Box Plot)

5. Box plot of calories per plan



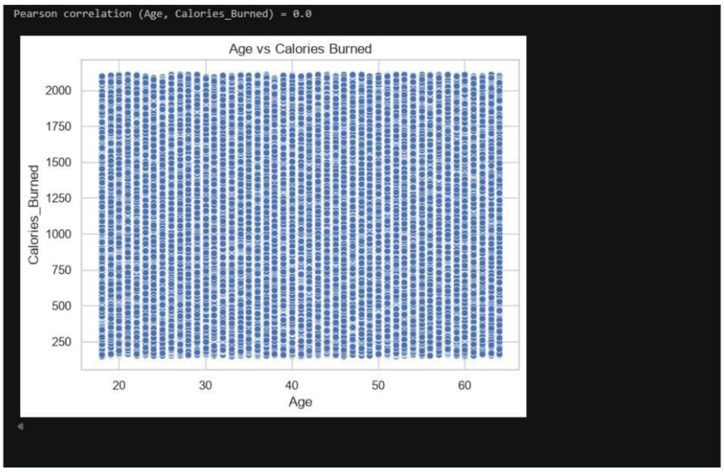
Monthly Sign-ups Over Time

6. Monthly sign-ups line chart



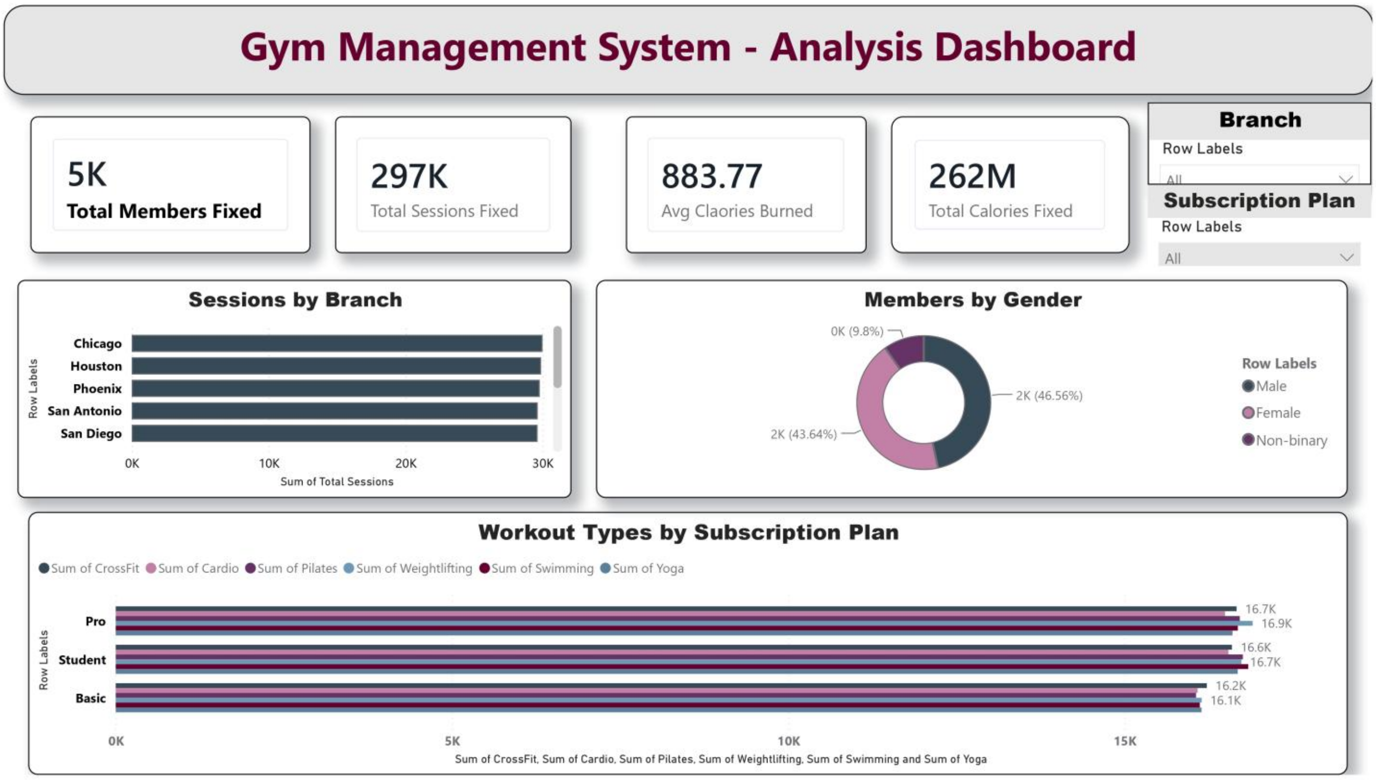
Correlation Between Age and Calories Burned

7. Correlation between age and calories



KPIs Dashboard

KPI	Value
Total Members	5,000
Total Sessions	~297,000
Avg. Calories Burned	883.77
Total Calories Burned	262M



Power BI — Gym Management System Analysis Dashboard

Conclusion

- The gym has a total of 5,000 members, distributed relatively evenly across 5 branches.
- Approximately 297,000 clean workout sessions were recorded after fully removing duplicates and missing values.
- No statistical relationship was found between member age and calories burned, and no material differences were observed across subscription plans.
- Session data quality improved from 305K rows with multiple issues to a fully clean set of 297K rows after the three-stage cleaning process.

Deliverables

- Raw data files
- Cleaning files
- Analysis files
- Power BI file
- Documentation file (this document)

Thank You

We sincerely thank the Ministry of Communications and Information Technology for their continuous guidance and support.

We also extend our sincere appreciation to our supervisor, Eng. Menna Tarek, for her valuable guidance and encouragement throughout this project.